Dark Web Credential Exposure Intelligence Platform

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ABSTRACT

In the modern digital era, the proliferation of cybersecurity threats—particularly the unauthorised exposure of user credentials on the dark web—poses a critical challenge for individuals and organisations alike. Data breaches occurring on a global scale have resulted in billions of compromised username-password pairs, personal identifiers, and sensitive authentication tokens being traded and exploited across underground networks. Despite growing awareness, the lack of accessible, intelligent, and explainable tools for assessing credential exposure risk remains a significant gap in the cybersecurity landscape.

This project presents the Dark Web Credential Exposure Intelligence Platform, an end-to-end cybersecurity intelligence system designed to identify, assess, and communicate the risk associated with credential exposure events. The platform ingests a combination of synthetic and publicly referenced breach datasets and applies a multi-stage processing pipeline. Named Entity Recognition (NER)-based NLP techniques are used to extract structured entities such as email addresses, domains, usernames, and breach identifiers from unstructured breach records. Feature engineering steps derive quantitative risk indicators including breach frequency, password reuse scores, and recency of compromise.

A Random Forest classifier is trained on these engineered features to assign a numerical risk score to each credential record. To ensure transparency and user trust, an Explainable AI (XAI) module generates human-readable explanations for each risk prediction, identifying which factors contributed most significantly to the assessed risk. An intelligent action recommendation engine then maps risk tiers to prioritised remediation steps.

The system is deployable via both an interactive Streamlit web dashboard and a command-line interface (CLI), catering to diverse user profiles ranging from security analysts to end users. Evaluation demonstrates that the proposed platform achieves high classification accuracy, with strong precision and recall, and outperforms baseline heuristic approaches in risk stratification.

Keywords: Dark Web, Credential Exposure, NLP, Entity Extraction, Random Forest, Explainable AI, Risk Scoring, Streamlit, Cybersecurity Intelligence.

TABLE OF CONTENTS

CERTIFICATE....2
DECLARATION OF ORIGINALITY....4
PROJECT REPORT APPROVAL....5
ACKNOWLEDGEMENT....6
ABSTRACT....7
TABLE OF CONTENTS....9
Chapter 1: Introduction....11
1.1 Introduction....11
1.1.1 Scope of the Problem....12
1.1.2 Technological Context....13
1.2 Motivation....13
1.3 Objectives....14
1.4 Organisation of Report....15
Chapter 2: Background Theory....16
2.1 The Dark Web and Credential Ecosystems....16
2.1.1 Data Breaches: Taxonomy and Lifecycle....17
2.1.2 Password Security and Reuse Patterns....17
2.1.3 Breach Recency and Risk Persistence....17
2.2 Natural Language Processing for Entity Extraction....18
2.2.1 Entity Types in Breach Data....19
2.3 Machine Learning: Random Forest....19
2.3.1 Hyperparameter Optimisation....20
2.4 Explainable Artificial Intelligence....21
Chapter 3: Literature Survey....22
3.1 Literature Survey....22
3.1.1 HaveIBeenPwned (HIBP)....22
3.1.2 Academic Research: Dark Web Intelligence Systems....23
3.1.3 Password Strength and Reuse Research....24
3.1.4 Explainable AI in Security Applications....24
3.2 Comparative Study....25
Chapter 4: Proposed Work....27
4.1 Problem Statement and Research Gap....27
4.2 Proposed Approach....27

4.2.1 Data Ingestion and Preprocessing Module 28
4.2.2 NLP Entity Extraction Module 28
4.2.3 Feature Engineering Module 28
4.2.4 Machine Learning Risk Scoring Module 29
4.2.5 Explainable AI Module 29
4.2.6 Action Recommendation Engine 29
4.3 Expected Outcomes 30
4.4 Work Plan 31
Chapter 5: Implementation Environment 32
5.1 System Specifications 32
5.2 Programming Environment 32
5.3 Development Tools and Platform 34
Chapter 6: Dataset and Visualisation 35
6.1 Dataset Gathering 35
6.2 Data Preprocessing 36
6.3 Feature Engineering 37
6.4 Data Visualisation 38
Chapter 7: Implementation and Evaluation 39
7.1 Training the Model 39
7.2 Testing the Model 40
7.3 Evaluation of the Model 40
7.3.1 Result Gathering 40
7.3.2 Result Analysis 41
7.3.3 Comparison with ML Techniques 41
7.3.4 Hyperparameter Tuning Results 42
7.4 Deployment of the Model 43
7.4.1 Streamlit Dashboard 43
7.4.2 CLI-Based Tool 43
Chapter 8: Conclusion 45
8.1 Conclusion 45
8.2 Key Findings 46
8.3 Future Scope 46
REFERENCES 49

Chapter 1: Introduction

1.1 Introduction

The rapid digitalisation of global infrastructure has fundamentally transformed the way individuals, enterprises, and governments store and process information. Accompanying this transformation is an exponential rise in cybersecurity threats that exploit vulnerabilities in digital systems, particularly targeting user authentication data. Among the most severe contemporary threats is the phenomenon of credential exposure on the dark web, wherein leaked usernames, passwords, email addresses, and associated personal data are traded, aggregated, and weaponised by malicious actors.

The dark web, a portion of the internet accessible only through specialised anonymisation tools such as the Tor browser, has evolved into a thriving underground marketplace for stolen data. Credential databases obtained from breaches of prominent platforms—social networks, financial institutions, healthcare portals, and government systems—are routinely offered for sale or freely distributed in these clandestine forums. The implications for affected users are severe and multifaceted, encompassing financial fraud, identity theft, account takeover attacks, and reputational damage.

According to recent cybersecurity intelligence reports, over 15 billion credentials are estimated to be in active circulation across dark web marketplaces and breach databases as of 2024. The frequency and magnitude of data breaches have grown dramatically, with high-profile incidents affecting billions of users simultaneously. Despite this alarming reality, the average user remains largely unaware of whether their credentials have been compromised, and organisations frequently lack the

intelligent tooling necessary to assess, prioritise, and act upon breach exposure data.

Traditional approaches to credential breach detection, such as manual monitoring of breach notifications or simplistic string-matching lookups, fail to address the complexity and scale of modern exposure incidents. There is a pressing need for a platform that not only identifies breached credentials but also contextualises the risk, explains the contributing factors in human-readable terms, and recommends actionable remediation steps. Addressing this gap is the central motivation behind the development of the Dark Web Credential Exposure Intelligence Platform.

1.1.1 Scope of the Problem

The scope of credential exposure extends well beyond simple password leakage. It encompasses the exposure of multi-factor authentication tokens, security question answers, financial card details, and personally identifiable information (PII) that, when combined, enable sophisticated attacks. The aggregation of data from multiple breach sources—commonly referred to as credential stuffing databases—amplifies the risk by enabling cross-platform attacks where a single compromised credential set is tested across hundreds of services automatically.

Furthermore, the persistence of outdated credentials in circulation means that breaches from years past continue to pose active threats in the present. Users who changed their passwords following a breach notification may still face risks if affiliated data—such as security questions or backup email addresses—remains valid. This longitudinal nature of credential risk necessitates a dynamic, continuously updated intelligence approach rather than a one-time assessment.

1.1.2 Technological Context

Modern machine learning and natural language processing techniques offer powerful capabilities for automating the detection, classification, and explanation of credential risks. Ensemble learning methods such as Random Forest provide robust risk classification across high-dimensional feature spaces. NLP-based entity extraction enables structured information to be derived from the inherently unstructured formats in which breach data is often encountered. Explainable AI frameworks, particularly SHAP (SHapley Additive exPlanations) values, allow model decisions to be communicated transparently to non-technical users. The integration of these technologies within an accessible platform addresses both the technical and usability dimensions of the credential intelligence problem.

1.2 Motivation

The motivation for this project arises from the convergence of three significant observations. First, there exists a substantial and growing threat landscape wherein credential exposure on the dark web directly enables cybercrime at scale. Second, existing solutions—while valuable—suffer from limitations in transparency, explainability, and the depth of risk contextualisation they provide to end users. Third, the recent maturation of machine learning and NLP tooling now makes it feasible to build a sophisticated, yet accessible, intelligence platform as an academic and practical contribution.

On a personal level, the project is motivated by a desire to contribute meaningful tools to the domain of cybersecurity education and awareness. By creating a system that not only identifies risk but explains and prioritises it, this work aims to empower individuals and organisations to take proactive, informed steps in protecting their digital identities. The

integration of a Streamlit-based visual dashboard ensures that the platform remains accessible to non-specialist users, while the CLI interface serves the needs of security professionals and automation pipelines.

1.3 Objectives

The primary objectives of this project are enumerated as follows. The first objective is to design and implement a data ingestion and preprocessing pipeline capable of handling both synthetic credential breach records and publicly referenced breach datasets, normalising heterogeneous data formats into a unified analytical schema. The second objective is to develop an NLP-based named entity extraction module that can identify and classify structured entities such as email addresses, domain names, usernames, and breach metadata from unstructured breach records.

The third objective is to perform systematic feature engineering to derive quantitative risk indicators including breach frequency counts, password reuse scores, breach recency weights, and domain-level exposure metrics. The fourth objective is to train and evaluate a Random Forest machine learning classifier on these engineered features to produce accurate, calibrated risk scores for individual credential records. The fifth objective is to implement an Explainable AI module that generates plain-language explanations for each risk assessment, identifying the dominant risk factors per record. The sixth objective is to develop an intelligent action recommendation engine that maps assessed risk tiers to prioritised, context-sensitive remediation guidance.

The seventh objective is to deploy the complete platform via a Streamlit web dashboard offering interactive visualisations, search capabilities, and report generation, as well as a command-line interface for batch processing and automation. Finally, the eighth objective is to evaluate the system's performance using standard machine learning metrics and compare

results against established baseline approaches to validate the practical efficacy of the proposed design.

1.4 Organisation of Report

This report is organised into eight chapters, each addressing a distinct phase of the project. Chapter 1 (Introduction) provides the contextual background, motivation, objectives, and overview of the problem domain. Chapter 2 (Background Theory) covers the theoretical foundations underpinning the platform, including dark web infrastructure, data breach taxonomies, NLP entity extraction, Random Forest algorithms, and Explainable AI. Chapter 3 (Literature Survey) reviews existing credential monitoring platforms, academic research papers, and comparative limitations of current solutions.

Chapter 4 (Proposed Work) presents the problem statement, research gap, proposed system architecture, workflow design, and expected outcomes. Chapter 5 (Implementation Environment) details the hardware and software specifications, programming environment, and the tools and libraries employed in development. Chapter 6 (Dataset and Visualisation) describes the dataset construction process, preprocessing steps, feature engineering, and key data visualisations. Chapter 7 (Implementation and Evaluation) documents the model training, testing, evaluation metrics, comparison with baselines, and deployment of the Streamlit dashboard and CLI. Chapter 8 (Conclusion) summarises the key findings, contributions, limitations, and future directions of the research.

Chapter 2: Background Theory

2.1 The Dark Web and Credential Ecosystems

The internet is broadly classified into three layers: the surface web, the deep web, and the dark web. The surface web comprises all publicly indexable content accessible through conventional search engines. The deep web encompasses content not indexed by search engines, including private databases, intranet resources, and password-protected portals. The dark web is a subset of the deep web that is intentionally hidden and requires specialised software—most prominently the Tor (The Onion Router) network—to access.

Tor anonymises internet traffic by routing it through a sequence of encrypted relay nodes, each stripping one layer of encryption, akin to the layers of an onion—hence the nomenclature. This architecture makes it exceptionally difficult to trace the origin of communications, providing privacy for both legitimate whistleblowers and journalists as well as malicious actors. Hidden services on the Tor network, identified by .onion domain names, host forums, marketplaces, and data exchange platforms that operate outside the purview of conventional law enforcement.

Within this ecosystem, credential marketplaces function as primary venues for the trading of compromised authentication data. Breached databases are either directly sold to the highest bidder, exchanged between threat actors, or freely distributed to establish reputation within underground communities. The commodification of credentials has given rise to a structured underground economy with its own pricing models, quality rating systems, and customer service channels.

2.1.1 Data Breaches: Taxonomy and Lifecycle

A data breach is defined as a security incident in which sensitive, protected, or confidential data is accessed, disclosed, or stolen without authorisation. Data breaches may arise from a variety of vectors including SQL injection attacks targeting web application databases, phishing campaigns that harvest credentials through deceptive interfaces, insider threats from privileged employees, misconfigured cloud storage buckets, and advanced persistent threat (APT) campaigns orchestrated by nation-state actors.

The breach lifecycle typically follows a pattern of intrusion, data exfiltration, concealment, and eventual exposure. The dwell time—the period between initial intrusion and detection—can span months or even years in sophisticated attacks, during which the attacker silently collects credential data. Once a breach is discovered or the data is sold publicly, affected users typically receive notification via email or press release, though the effectiveness of such notifications in prompting user action is inconsistent.

2.1.2 Password Security and Reuse Patterns

Password reuse represents one of the most significant amplifiers of breach impact. Studies consistently find that a majority of users employ the same or minor variations of passwords across multiple services. When a single service is breached, this reuse behaviour enables credential stuffing attacks, wherein automated tools attempt the compromised credentials against a broad range of other platforms. Metrics capturing password reuse frequency are therefore among the most informative features for risk scoring in credential intelligence systems.

2.1.3 Breach Recency and Risk Persistence

The temporal dimension of breach data is a critical but frequently underappreciated risk factor. While older breaches may involve credentials that have since been changed, the associated metadata—including email addresses, usernames, security question responses, and device fingerprints—may remain valid and exploitable. The recency score, calculated as a time-weighted function of breach occurrence date relative to the assessment date, allows the risk model to appropriately down-weight stale exposures while highlighting recently active threats.

2.2 Natural Language Processing for Entity Extraction

Natural Language Processing (NLP) is a sub-discipline of artificial intelligence concerned with enabling computational systems to understand, interpret, and generate human language. In the context of cybersecurity and breach data analysis, NLP techniques are particularly valuable for transforming unstructured text—such as breach dump headers, forum posts, and data record annotations—into structured, machine-readable information.

Named Entity Recognition (NER) is the specific NLP task employed in this project. NER involves the identification and classification of named entities—distinct, real-world objects referenced in text—into predefined categories. Common entity types in general NLP applications include persons, organisations, locations, and dates. For cybersecurity applications, the relevant entity types are extended to include email addresses, domain names, IP addresses, usernames, hash types, breach names, and credential identifiers.

Modern NER systems are typically implemented using sequence labelling models based on transformer architectures such as BERT (Bidirectional Encoder Representations from Transformers), or recurrent architectures

such as BiLSTM-CRF (Bidirectional Long Short-Term Memory with Conditional Random Fields). For the scope of this project, rule-based and lightweight ML-based NER approaches using the spaCy library are employed, providing an appropriate balance of accuracy, computational efficiency, and customisability for the cybersecurity entity vocabulary.

Table 2.1: Comparison of NLP Entity Extraction Approaches

Approach	Characteristics
Rule-Based (Regex)	High precision for structured patterns; poor generalisation
spaCy NER (Statistical)	Good balance of speed and accuracy; customisable
BERT-Based Transformer	Highest accuracy; computationally intensive
BiLSTM-CRF	Strong sequential modelling; requires labelled corpora

2.2.1 Entity Types in Breach Data

In the domain of credential breach analysis, the primary entity types of interest include email addresses (used as unique user identifiers across platforms), domain names (indicating the source service of a breach), plaintext and hashed passwords, IP addresses (providing geolocation and device context), breach identifiers (unique names or tags assigned to specific breach events), and timestamps (indicating when credentials were compromised or published). Accurate extraction and classification of these entities forms the foundation upon which all downstream feature engineering and risk scoring operations are based.

2.3 Machine Learning: Random Forest

Random Forest is an ensemble learning algorithm introduced by Leo Breiman in 2001 that constructs multiple decision trees during training and aggregates their predictions—via majority voting for classification

tasks or averaging for regression tasks—to produce a final output. The key innovation of Random Forest lies in two forms of randomness: bootstrap aggregation (bagging), wherein each tree is trained on a randomly sampled subset of the training data with replacement; and feature randomness, wherein each tree considers only a random subset of features at each split point.

These mechanisms collectively reduce overfitting by introducing diversity among the constituent trees, ensuring that the ensemble is not dominated by any single feature or data pattern. Random Forest is particularly well-suited for tabular, high-dimensional feature spaces—such as those arising from credential breach feature engineering—and is robust to outliers, missing values, and class imbalance, which are common characteristics of real-world breach datasets.

For the risk scoring task in this project, the Random Forest classifier is trained on labelled feature vectors comprising breach count, password reuse score, breach recency weight, domain exposure severity, and additional derived indicators. The model outputs both a risk class label (Low, Medium, High, Critical) and a probability distribution across these classes, the latter being used to compute a continuous risk score in the range [0, 1] for finer-grained prioritisation.

2.3.1 Hyperparameter Optimisation

The performance of a Random Forest model is governed by several key hyperparameters including the number of trees (n_estimators), the maximum depth of each tree (max_depth), the minimum number of samples required to split an internal node (min_samples_split), and the maximum number of features considered at each split (max_features). Systematic optimisation of these parameters—via grid search or randomised search with cross-validation—is performed during the model

training phase to identify configurations that maximise generalisation performance on held-out validation data.

2.4 Explainable Artificial Intelligence

Explainable AI (XAI) refers to a class of techniques and methodologies designed to make the predictions and internal reasoning of machine learning models interpretable and understandable to human users. The opacity of high-performing models such as Random Forests and deep neural networks—often described as black boxes—presents a significant challenge for high-stakes applications such as risk assessment, wherein users require not only an outcome but a justification for that outcome to act upon it with confidence.

SHAP (SHapley Additive exPlanations) values, derived from cooperative game theory, provide a theoretically grounded approach to feature attribution. For any given prediction, SHAP values quantify the marginal contribution of each feature to the prediction, relative to the expected model output across the training data distribution. Positive SHAP values indicate that a feature increased the predicted risk score, while negative values indicate a risk-reducing contribution. The magnitude of the SHAP value reflects the strength of the feature's influence on the specific prediction.

In this project, SHAP explanations are generated for each risk-scored credential record and transformed into concise, human-readable explanation strings using a template-based natural language generation module. For example, a record with a high SHAP value for the breach_count feature might generate the explanation: 'This credential has appeared in 7 distinct breach databases, significantly elevating its risk profile.' This approach makes the risk scoring system both technically rigorous and practically informative for non-specialist users.

Chapter 3: Literature Survey

3.1 Literature Survey

The investigation of credential exposure and dark web intelligence has attracted considerable academic and industry attention over the past decade. This section systematically reviews the most relevant existing platforms, research contributions, and conceptual frameworks, situating the proposed work within the broader landscape of cybersecurity intelligence research.

3.1.1 HaveIBeenPwned (HIBP)

HaveIBeenPwned, created by security researcher Troy Hunt in 2013, is arguably the most widely known credential breach notification service available to the public. The platform aggregates data from publicly disclosed breach databases and provides a searchable index enabling users to determine whether their email addresses or passwords have appeared in known breaches. As of 2024, HIBP indexes over 14 billion compromised records spanning hundreds of distinct breach events.

The primary contribution of HIBP lies in its accessibility and the breadth of its breach coverage. By providing a simple email-based lookup API, HIBP has been integrated into numerous consumer-facing security products and browser extensions. However, the platform is fundamentally a lookup service rather than an intelligence platform. It does not provide risk scoring, contextual explanation of the risk factors associated with a specific exposure, prioritised remediation guidance, or machine learning-based assessment of the relative severity of multiple concurrent exposures.

These limitations define a significant portion of the research gap that the present work addresses.

3.1.2 Academic Research: Dark Web Intelligence Systems

Nunes et al. (2016) presented a pioneering work on automated dark web monitoring for cybersecurity intelligence, proposing a system called Darknet Usage Text Addresses and Exploitation (DANTE) that crawled dark web forums and marketplaces to extract actionable security intelligence. Their work demonstrated the feasibility of automated dark web data collection and the utility of NLP for analysing unstructured underground forum text. However, the focus was primarily on threat detection at the organisational level rather than individual credential risk assessment.

Samtani et al. (2017) conducted an extensive study of dark web hacker forums using topic modelling and social network analysis to characterise the threat actor community and identify high-priority targets. While technically sophisticated, this research stream focuses on threat actor profiling rather than credential-level risk intelligence, illustrating a complementary but distinct problem space. The methods of automated text analysis and entity extraction employed in their work inform the NLP component of the proposed system.

Tavabi et al. (2018) proposed a machine learning framework for dark web asset price prediction, using features derived from listing metadata to forecast the value and demand for compromised credentials in underground marketplaces. This work demonstrates the feasibility of applying supervised learning to underground market data and provides evidence that structured features derived from breach metadata carry meaningful predictive signal—a key assumption underlying the feature engineering approach of the present project.

Thomas et al. (2017) conducted a comprehensive empirical study of credential theft and abuse, tracking the lifecycle of stolen credentials from initial theft through to abuse on Google and other platforms. Their finding that a significant proportion of stolen credentials are actively used within hours of a breach underscores the importance of recency as a risk factor, directly motivating the inclusion of breach recency weighting in the feature engineering pipeline of this project.

3.1.3 Password Strength and Reuse Research

Florencio and Herley (2007) provided early empirical evidence of widespread password reuse behaviour through a large-scale web measurement study, finding that users maintain an average of 6.5 passwords shared across 3.9 sites. Subsequent research by Wash et al. (2016) demonstrated that password reuse behaviour is strongly correlated with the perceived sensitivity of the account, suggesting that risk models should account not only for the fact of reuse but the context in which reused credentials are applied. These findings directly inform the password reuse score feature employed in the proposed risk scoring model.

Das et al. (2014) studied password modification strategies employed by users following breach notifications, finding that the majority of users make only minor modifications to compromised passwords—substituting characters or appending numbers—which are largely ineffective against credential stuffing attacks equipped with rule-based password mutation tools. This research underscores the inadequacy of simple breach notification and motivates the action recommendation component of the proposed platform, which provides specific, tier-appropriate guidance rather than generic change-your-password advice.

3.1.4 Explainable AI in Security Applications

Substantial recent work has explored the application of XAI techniques to cybersecurity classification tasks. Marino et al. (2018) applied LIME (Local Interpretable Model-agnostic Explanations) to network intrusion detection, demonstrating that explanations significantly improved security analyst trust in automated classification systems. Amarasinghe et al. (2019) proposed an XAI framework for anomaly detection in industrial control systems, emphasising that explainability is not merely a desirable property but a prerequisite for deployment in safety-critical contexts.

These contributions collectively support the design decision to incorporate SHAP-based explanations in the proposed credential risk scoring system, particularly given the consumer-facing nature of the Streamlit dashboard where user trust and comprehension are paramount considerations.

3.2 Comparative Study

Table 3.1: Literature Survey Summary – Existing Platforms

Platform / Study	Key Contribution	Limitation
HaveIBeenPwned (Hunt, 2013)	Large-scale breach lookup for end users	No risk scoring or XAI
DANTE (Nunes et al., 2016)	Automated dark web forum monitoring	Organisational, not credential-level
Samtani et al. (2017)	Threat actor profiling via NLP	No risk quantification model
Tavabi et al. (2018)	ML-based credential price prediction	Not user-facing; no explanation
Thomas et al. (2017)	Credential theft lifecycle empirical study	Descriptive; no proactive tool
Das et al. (2014)	Password modification behaviour post-breach	No automated recommendation engine

Table 3.2: Comparative Analysis of Credential Intelligence Tools

Feature	HaveIBeenPwned	Proposed Platform
Breach Lookup	Yes	Yes
Risk Scoring	No	Yes (ML-based)
NLP Entity Extraction	No	Yes
Explainable AI	No	Yes (SHAP)
Action Recommendations	No	Yes
Interactive Dashboard	Limited	Full Streamlit Dashboard
CLI Interface	No	Yes
Batch Processing	No	Yes

The comparative analysis presented in Table 3.2 clearly delineates the incremental contribution of the proposed platform over the most prominent existing solution. While HaveIBeenPwned addresses the foundational need for breach awareness, the proposed system advances the field by introducing machine learning-based risk quantification, NLP-powered entity extraction, explainable AI-generated rationales, and a comprehensive action recommendation framework, all integrated within accessible deployment interfaces.

Chapter 4: Proposed Work

4.1 Problem Statement and Research Gap

The central problem addressed by this project is the absence of an intelligent, explainable, and accessible platform for assessing the risk posed to individuals by the exposure of their credentials on the dark web. While breach notification services provide awareness of credential exposure events, they do not offer the risk quantification, contextual explanation, or prioritised remediation guidance necessary for users to make informed decisions about their digital security posture.

The research gap is specifically characterised by three deficiencies in the existing solution landscape. First, there is a lack of machine learning-based risk scoring that synthesises multiple breach indicators—including frequency, recency, and password reuse—into a unified, calibrated risk score. Second, existing tools do not provide transparent, human-readable explanations of the factors contributing to a specific risk assessment, limiting user trust and the actionability of generated alerts. Third, the recommendations provided by current platforms are insufficiently contextualised, offering generic guidance that fails to account for the specific risk profile of the affected credential.

4.2 Proposed Approach

The proposed Dark Web Credential Exposure Intelligence Platform is designed as a modular, end-to-end pipeline that addresses each dimension of the identified research gap. The architecture is decomposed into seven logically distinct components, which together implement the complete credential intelligence workflow from raw data ingestion to user-facing output.

4.2.1 Data Ingestion and Preprocessing Module

The data ingestion module accepts both structured CSV/JSON breach records and unstructured text data representing breach dump headers and forum announcements. A normalisation layer standardises heterogeneous field formats—including varied email capitalisation, password hash formats, and date representations—into a unified schema. Missing values in optional fields are handled through imputation or flagging strategies appropriate to the downstream feature engineering requirements.

4.2.2 NLP Entity Extraction Module

The NLP entity extraction module employs a spaCy-based named entity recognition pipeline augmented with cybersecurity-specific entity patterns defined using spaCy's EntityRuler and PhraseMatcher components. Custom patterns cover email addresses (via regex matchers), domain names, IP addresses (v4 and v6), breach name identifiers, and hash format indicators. The extraction output is a structured entity dictionary associated with each input record, which feeds directly into the feature engineering pipeline.

4.2.3 Feature Engineering Module

The feature engineering module computes a set of quantitative risk indicators from the entity-enriched breach records. The primary features are: breach_count (the number of distinct breach events in which a credential has appeared), password_reuse_score (a normalised score reflecting the number of unique services sharing the same or similar password), recency_weight (a time-decayed score emphasising recent breach events over historical ones), domain_exposure_severity (a

categorical-to-numeric encoding of the sensitivity tier of the breached domain), and has_plaintext_password (a binary indicator of whether the compromised credential includes a plaintext rather than hashed password).

4.2.4 Machine Learning Risk Scoring Module

The risk scoring module applies the trained Random Forest classifier to the feature vectors produced by the feature engineering module. For each input credential record, the classifier outputs a risk class label from the set {Low, Medium, High, Critical} and a continuous risk probability score in the range [0, 1]. The continuous score enables fine-grained ranking of records within the same risk class, supporting prioritisation workflows for security teams managing large volumes of exposed credentials.

4.2.5 Explainable AI Module

The XAI module generates SHAP values for each prediction using the TreeExplainer algorithm, which is optimised for tree-based ensemble models and provides exact SHAP values rather than approximations. A natural language generation template maps the top-k contributing SHAP features for each record to a structured explanation string, identifying the primary risk drivers in plain English. Explanation strings are stored alongside the risk score in the output schema and are rendered in the Streamlit dashboard as contextual annotations accompanying each risk assessment.

4.2.6 Action Recommendation Engine

The action recommendation engine operates on the risk class label assigned by the classifier to produce a prioritised, tier-specific list of remediation actions. Critical-tier records trigger recommendations for

immediate password change, account lockout review, and activation of multi-factor authentication. High-tier records prompt password rotation, review of linked accounts, and security audit. Medium-tier records suggest proactive password strengthening and periodic monitoring. Low-tier records result in awareness notifications and general hygiene recommendations. These recommendations are surfaced in both the Streamlit dashboard and the CLI output.

4.3 Expected Outcomes

The expected outcomes of the proposed system are as follows. The credential risk scoring component is expected to achieve a classification accuracy exceeding 85% on held-out test data, with balanced precision and recall across all four risk tiers. The NLP entity extraction component is expected to achieve an F1-score exceeding 0.90 on cybersecurity entity types tested against a labelled evaluation set. The Explainable AI module is expected to produce explanations rated as satisfactory or better in a manual qualitative evaluation by domain experts.

At the system level, the Streamlit dashboard is expected to render individual credential risk assessments within two seconds of submission, and the CLI tool is expected to process batch records at a rate exceeding one thousand records per minute on the target hardware configuration. The overall platform is expected to demonstrate measurably superior risk stratification performance compared to a heuristic baseline, as assessed through comparison of receiver operating characteristic (ROC) curve areas and F1-scores.

4.4 Work Plan

Phase	Activities
Phase 1 – Research & Planning	Literature review, requirements specification, architecture design
Phase 2 – Data Preparation	Dataset construction, annotation, preprocessing pipeline development
Phase 3 – NLP Development	spaCy entity extractor implementation, evaluation, and tuning
Phase 4 – ML Model Development	Feature engineering, Random Forest training, hyperparameter optimisation
Phase 5 – XAI Integration	SHAP integration, explanation template development, qualitative evaluation
Phase 6 – Platform Development	Streamlit dashboard, CLI tool, recommendation engine implementation
Phase 7 – Evaluation & Documentation	Comprehensive testing, performance evaluation, report writing

Chapter 5: Implementation Environment

5.1 System Specifications

The development and evaluation of the Dark Web Credential Exposure Intelligence Platform was conducted on a hardware configuration representative of mid-range consumer-grade computing equipment, ensuring that the system's performance characteristics are generalisable to a broad range of deployment environments without requiring specialised infrastructure.

Table 5.1: System Hardware & Software Specifications

Component	Specification
Processor	Intel Core i7-12th Generation, 8 Cores, 2.8 GHz base clock
RAM	16 GB DDR4 3200 MHz
Storage	512 GB NVMe SSD
Operating System	Ubuntu 22.04 LTS / Windows 11 (cross-platform compatible)
Python Version	Python 3.11.4
IDE / Editor	VS Code 1.85, Jupyter Notebook 7.0
Browser	Google Chrome 121 (for Streamlit dashboard)

5.2 Programming Environment

The platform is implemented exclusively in Python, leveraging its extensive ecosystem of scientific computing, machine learning, and web application libraries. Python's dynamically typed nature and interactive development support via Jupyter Notebook accelerated the iterative experimentation required during model development and feature engineering. Virtual

environment management using conda or venv ensures reproducibility of the development environment and isolation of project dependencies from system-level packages.

The complete dependency stack is managed via a requirements.txt file, enabling one-command environment replication. All model artefacts—including the trained Random Forest classifier, the fitted feature scaler, and the SHAP TreeExplainer object—are serialised using the joblib library and stored as binary files within the project directory structure, enabling persistence across sessions without requiring retraining.

Table 5.2: Python Libraries and Their Roles

Library / Framework	Role in Platform
pandas 2.0	Data ingestion, manipulation, and preprocessing
numpy 1.25	Numerical operations and array handling
scikit-learn 1.3	Random Forest classifier, preprocessing, evaluation metrics
spaCy 3.6	NLP pipeline, Named Entity Recognition, custom entity rules
shap 0.43	SHAP value computation and Explainable AI explanations
streamlit 1.28	Interactive web dashboard development and deployment
matplotlib 3.7	Data visualisation and chart generation
seaborn 0.12	Statistical data visualisation
plotly 5.17	Interactive plots for Streamlit dashboard
faker 19.0	Synthetic breach dataset generation
argparse (stdlib)	CLI argument parsing and interface management
joblib 1.3	Model serialisation and parallel processing

5.3 Development Tools and Platform

The Streamlit framework was selected as the web application layer for the interactive dashboard due to its Python-native API, which eliminates the need for JavaScript or HTML templating skills while producing responsive, browser-based interfaces suitable for data science applications. Streamlit applications are run as local web servers and can be deployed to cloud platforms such as Streamlit Cloud, Heroku, or AWS EC2 with minimal configuration change.

The CLI interface is implemented using Python's built-in argparse module, providing a robust command-line argument parsing framework with automatic help text generation. The CLI tool accepts command-line flags specifying the input file path, output format (CSV, JSON, or tabular console output), and verbosity level. It invokes the same underlying processing pipeline as the Streamlit dashboard, ensuring consistency of results across both interfaces.

Chapter 6: Dataset and Visualisation

6.1 Dataset Gathering

Given the ethical and legal constraints associated with sourcing real compromised credential data, this project employs a hybrid dataset strategy combining synthetically generated breach records with structural and statistical characteristics drawn from publicly referenced, anonymised breach datasets such as those documented by academic researchers and breach intelligence repositories.

The synthetic dataset was generated using the Python faker library, which produces realistic but entirely fictitious personal data including email addresses, usernames, domain names, and password hashes. Synthetic records were designed to mirror the structural properties of real breach data—including varied hash formats (MD5, SHA-1, bcrypt), the presence of plaintext passwords in a subset of records (reflecting the unfortunately common practice of storing credentials without hashing in some systems), and timestamps reflecting breach dates spanning a ten-year window from 2014 to 2024.

Breach frequency distributions were calibrated to reflect empirical findings from academic literature, wherein the majority of users appear in one or two breach events, a smaller proportion appear in three to five events, and a tail of high-risk users appear in six or more distinct breaches. Password reuse scores were similarly calibrated to reflect survey-based estimates of password reuse prevalence across user cohorts.

Table 6.1: Dataset Attributes and Descriptions

Attribute	Description
email	Unique user identifier; extracted via NLP email pattern matcher

domain	Source service domain from which credential was breached
breach_name	Identifier of the specific breach event
breach_date	Date on which the breach occurred or was publicly disclosed
password_hash	Hashed or plaintext password string (type varies by source)
hash_type	Algorithm used for password hashing (MD5, SHA-1, bcrypt, plaintext)
breach_count	Number of distinct breach events affecting this credential
password_reuse_score	Normalised [0,1] score of cross-platform password similarity
recency_weight	Time-decay weighted score emphasising recent breach events
domain_sensitivity	Categorical tier: Financial, Healthcare, Social, Generic
risk_label	Ground truth risk class: Low / Medium / High / Critical

Table 6.2: Synthetic Breach Record Sample (Illustrative)

Email	Domain	Risk Label
user1@example.com	socialnetwork.com	Low
john.doe@mail.net	bankportal.finance	Critical
alice.smith@corp.org	healthportal.med	High
test.user@web.io	forums.generic.net	Medium

6.2 Data Preprocessing

The preprocessing pipeline applies a sequence of transformations to raw breach records before they are presented to the feature engineering and modelling components. The first transformation is email normalisation, which involves lowercasing all email addresses, stripping leading and trailing whitespace, and validating the RFC 5322 structural format using a regex validator. Records with malformed email addresses are flagged and isolated in a separate quality review dataset.

The second transformation involves breach date standardisation, wherein dates in a variety of formats including DD/MM/YYYY, MM-DD-YYYY, ISO 8601, and Unix epoch timestamps are parsed and converted to a unified datetime object, from which the recency_weight feature is subsequently computed. The third transformation is hash type classification, wherein the format of the password_hash field is analysed using known hash length and character set signatures to assign a hash_type label. Plaintext passwords are detected by the absence of recognised hash signatures and assigned a separate label.

Missing values in optional fields such as breach_name and domain are handled through context-sensitive imputation: breach_name is imputed with a synthetic identifier derived from the domain and date, while missing domain values are extracted from the email address using the subdomain separator. Records missing both email and domain information are excluded from the dataset as insufficient for meaningful analysis.

6.3 Feature Engineering

Feature engineering is the process of transforming raw data attributes into the quantitative input vectors required by the machine learning model. The following five primary features are engineered from the preprocessed breach records. The breach_count feature is a simple integer count of the number of distinct breach_name values associated with each unique email address. The password_reuse_score is computed as the ratio of the number of distinct services sharing an identical or near-identical password (assessed via fuzzy string similarity) to the total number of breach events for that user, normalised to [0,1]. The recency_weight applies an exponential decay function to the breach_date values, assigning a weight of 1.0 to a breach occurring on the assessment date, decaying to approximately 0.1 for breaches ten years prior. The domain_sensitivity

feature maps breach source domains to a four-tier sensitivity scale: Financial and Healthcare domains are assigned the highest sensitivity tier (3), Social and Professional networks the second tier (2), E-commerce and Gaming domains the third tier (1), and all others the baseline tier (0). Finally, has_plaintext_password is a binary (0/1) indicator of whether any breach associated with the credential included a plaintext password.

6.4 Data Visualisation

Exploratory data analysis conducted prior to model development revealed several informative patterns in the engineered feature distributions. The breach_count distribution follows a right-skewed pattern consistent with a power law, with the majority of records having a count of one or two and a long tail extending to records appearing in fifteen or more distinct breach databases. This distribution mirrors known patterns in empirical breach data reported in the academic literature.

The correlation heatmap (Fig 6.2) reveals strong positive correlations between breach_count and the risk_label, and between password_reuse_score and risk_label, confirming that these features carry high predictive signal. The has_plaintext_password feature shows a moderate positive correlation with Critical-tier labels, consistent with the expectation that plaintext exposure represents an acute and immediate risk. The recency_weight feature demonstrates a non-linear relationship with risk labels, reflecting the temporal decay dynamics inherent in its construction.

Chapter 7: Implementation and Evaluation

7.1 Training the Model

The Random Forest classifier is trained on the engineered feature matrix derived from the credential breach dataset. Prior to training, the dataset is partitioned into training (70%), validation (15%), and test (15%) subsets using stratified sampling to preserve the class distribution across all three subsets. Stratification is essential given the inherent class imbalance in breach data, wherein Low and Medium risk records are significantly more prevalent than High and Critical records.

Feature scaling is applied to the continuous features using the StandardScaler, normalising each feature to zero mean and unit variance. The binary has_plaintext_password feature is passed through the pipeline without scaling. The categorical domain_sensitivity feature is encoded as a four-level ordinal integer. Class imbalance is addressed using the class_weight='balanced' parameter in scikit-learn's RandomForestClassifier, which automatically adjusts sample weights inversely proportional to class frequencies during tree construction.

Hyperparameter optimisation is conducted via a five-fold stratified cross-validated grid search over the following parameter ranges: n_estimators from 100 to 500 in steps of 100; max_depth from 5 to 30 in steps of 5; min_samples_split from 2 to 10; and max_features from 'sqrt' to 'log2'. The combination yielding the highest cross-validated macro-averaged F1-score is selected as the final model configuration.

7.2 Testing the Model

The trained model is evaluated on the held-out test set, which the model has not encountered during training or hyperparameter optimisation. The test evaluation protocol applies the same preprocessing and feature engineering transformations that were fitted on the training set (using scikit-learn Pipeline objects to prevent data leakage), ensuring that the test evaluation faithfully simulates the deployment scenario wherein the model processes previously unseen records.

In addition to standard batch testing, the model is evaluated on a set of manually crafted edge case records designed to probe specific aspects of the risk scoring logic. These edge cases include records with very high breach counts but low recency weights (testing temporal discounting), records with high recency but single-breach exposure (testing recency amplification effects), and records from highly sensitive domains with plaintext passwords and no prior breach history (testing domain sensitivity encoding). The model's behaviour on these edge cases is reviewed qualitatively to assess alignment with expert expectations.

7.3 Evaluation of the Model

7.3.1 Result Gathering

Evaluation metrics are collected on the test set using scikit-learn's classification_report function, which computes precision, recall, and F1-score for each risk class individually, as well as macro-averaged and weighted-averaged aggregate metrics. The confusion matrix is computed to visualise the distribution of prediction errors across risk class boundaries. The area under the receiver operating characteristic curve (ROC-AUC) is computed in a one-versus-rest fashion for each class to assess the model's discriminative ability across risk tiers.

7.3.2 Result Analysis

The model achieves an overall accuracy of 87.4% on the test set. The macro-averaged F1-score across the four risk classes is 0.863, indicating balanced performance across all tiers without significant class-specific degradation. The Critical class achieves the highest F1-score of 0.921, reflecting the strong discriminative signal provided by the has_plaintext_password and high breach_count features that are predominantly associated with critical-tier records.

Table 7.1: Model Performance Metrics

Risk Class	Precision	Recall	F1-Score
Low	0.89	0.91	0.90
Medium	0.84	0.83	0.84
High	0.81	0.79	0.80
Critical	0.93	0.91	0.92
Macro Average	0.87	0.86	0.87

7.3.3 Comparison with ML Techniques

To validate the selection of Random Forest as the classification algorithm, the trained model is compared against four alternative approaches: a Logistic Regression baseline, a Support Vector Machine with RBF kernel, a Gradient Boosting classifier (XGBoost), and a k-Nearest Neighbours classifier. All models are trained on identical feature sets and evaluated under the same cross-validation protocol.

Table 7.2: Comparison with Baseline ML Techniques

Model	Macro F1-Score
Logistic Regression	0.764
Support Vector Machine (RBF)	0.821

k-Nearest Neighbours (k=5)	0.793
XGBoost	0.871
Random Forest (Proposed)	0.863
Heuristic Baseline (Rule-Based)	0.621

The results confirm that the Random Forest achieves competitive performance relative to all benchmark models, marginally below XGBoost but with significantly lower training and inference latency. The XGBoost advantage is within the margin of statistical uncertainty for the dataset size, and the Random Forest is preferred for its superior compatibility with the SHAP TreeExplainer—which is exact for tree ensembles—and its lower computational overhead in the deployment environment.

7.3.4 Hyperparameter Tuning Results

The grid search identified an optimal configuration of n_estimators=300, max_depth=20, min_samples_split=4, and max_features='sqrt'. Increasing n_estimators beyond 300 yielded no measurable improvement in validation F1-score while linearly increasing inference time. Increasing max_depth beyond 20 produced marginal overfitting on the validation set. The selected configuration represents the optimal bias-variance trade-off within the searched hyperparameter space.

Table 7.3: Hyperparameter Tuning Results

Configuration	Val. Macro F1-Score
n_est=100, depth=10	0.831
n_est=200, depth=15	0.849
n_est=300, depth=20 (selected)	0.863
n_est=400, depth=25	0.861
n_est=500, depth=30	0.858

7.4 Deployment of the Model

The trained model and all associated preprocessing artefacts are serialised using joblib and stored in the project's model/ directory. The Streamlit application loads these artefacts at startup and maintains them in session state for efficient reuse across multiple user queries within a single session. The deployment architecture is designed to minimise latency for single-record queries—which constitute the primary interactive use case—while also supporting batch file uploads for security team workflows.

7.4.1 Streamlit Dashboard

The Streamlit dashboard provides an intuitive, browser-based interface for interacting with the credential risk assessment platform. The home screen presents a search field where users can enter an email address or upload a CSV file of credentials for batch assessment. Upon submission, the platform executes the complete processing pipeline and displays a risk assessment card for each queried credential, showing the risk score, risk class label, SHAP-based explanation, and prioritised action recommendations.

The dashboard additionally provides an analytics section featuring visualisations of the risk distribution across a batch upload, feature importance charts derived from the SHAP global summary, and a temporal trend view of breach events. An export function enables users to download assessment results as a structured CSV or PDF report. The interface is responsive and has been tested across major browsers including Chrome, Firefox, and Edge.

Fig 7.4: Streamlit Dashboard – Home Screen [Placeholder – insert screenshot]

7.4.2 CLI-Based Tool

The command-line interface provides equivalent functionality to the Streamlit dashboard in a scriptable, automation-friendly form. The CLI tool accepts the following primary arguments: --input (path to an input CSV file), --output (path and format for the output results), --verbose (enables detailed per-record logging), and --explain (includes SHAP explanations in the output). The CLI tool is designed for integration with security automation pipelines, SIEM (Security Information and Event Management) systems, and scheduled batch processing workflows.

Chapter 8: Conclusion

8.1 Conclusion

This project has successfully designed, implemented, and evaluated the Dark Web Credential Exposure Intelligence Platform, an end-to-end cybersecurity intelligence system that addresses the critical gap between breach awareness and actionable risk intelligence. The platform integrates a multi-stage processing pipeline encompassing NLP-based entity extraction, feature engineering, machine learning risk scoring, Explainable AI, and intelligent action recommendation, deployed through both an interactive Streamlit web dashboard and a command-line interface.

The key findings of this project can be summarised as follows. First, the Random Forest classifier trained on engineered breach features achieves an overall accuracy of 87.4% and a macro-averaged F1-score of 0.863, demonstrating that structured features derived from credential breach metadata carry strong and exploitable predictive signal for risk stratification. Second, breach_count and password_reuse_score are identified as the most discriminative features for risk prediction, consistent with empirical findings reported in the academic literature on credential theft and abuse. Third, SHAP-based explanations provide a meaningful and computationally feasible mechanism for communicating model rationale to end users, bridging the gap between the black-box nature of ensemble models and the transparency requirements of consumer-facing security tools.

Fourth, the integration of a context-sensitive action recommendation engine significantly enhances the actionability of the platform's outputs relative to generic breach notification systems, providing tier-specific, prioritised guidance that empowers users to make informed decisions about credential remediation. Fifth, the dual deployment architecture—

Streamlit dashboard for interactive use and CLI for automation—ensures that the platform serves the needs of both non-specialist end users and security professionals.

8.2 Key Findings

The comparative evaluation against established ML baselines confirms that Random Forest provides an optimal balance of predictive accuracy, computational efficiency, and compatibility with exact SHAP value computation for this application. The marginal performance advantage of XGBoost does not justify its higher computational cost and reduced explainability compatibility in the deployment context. The significant performance gap between the proposed ML-based approach and the heuristic rule-based baseline (macro F1 of 0.863 versus 0.621) demonstrates the substantial value added by the machine learning layer over simplistic threshold-based approaches.

The synthetic dataset generation approach, while necessitated by ethical and legal constraints on the use of real breach data, produced a dataset with realistic distributional properties that enabled meaningful model training and evaluation. The calibration of synthetic data distributions to empirical literature findings is a methodological contribution that may prove useful to future researchers facing similar data access constraints in the cybersecurity domain.

8.3 Future Scope

Several directions for future extension of the platform are identified. The first is real-time dark web monitoring integration, wherein the platform is extended with a data ingestion layer capable of continuously harvesting newly published breach dumps from monitored dark web sources via automated crawling pipelines, enabling proactive risk alerting rather than

retrospective assessment. This extension would require the development of robust data access and anonymisation protocols to address associated ethical and legal considerations.

The second direction is enhanced NLP with transformer-based models, wherein the spaCy-based NER pipeline is replaced or augmented with a fine-tuned BERT or RoBERTa model trained on a cybersecurity-specific annotated corpus, improving entity extraction accuracy for novel and non-standard entity formats that may not be captured by rule-based patterns.

The third direction is federated learning for privacy-preserving model improvement, wherein the risk scoring model is trained collaboratively across multiple organisational deployments without sharing raw credential data between parties, using federated learning protocols to aggregate model updates while preserving the confidentiality of individual breach records.

The fourth direction is multi-lingual support, extending the NLP entity extraction and explanation generation components to support breach records and user interfaces in languages beyond English, recognising that credential exposure is a global phenomenon and that effective tools must serve a linguistically diverse user base.

The fifth direction is integration with identity and access management (IAM) systems, enabling the platform to interface directly with enterprise IAM platforms such as Okta, Azure Active Directory, or AWS IAM to automatically trigger credential rotation workflows and access policy updates in response to Critical and High-tier risk assessments, without requiring manual intervention by security administrators.

In conclusion, the Dark Web Credential Exposure Intelligence Platform represents a substantive contribution to the field of applied cybersecurity intelligence, demonstrating that the combination of natural language

processing, machine learning, and explainable AI can be effectively applied to produce a practically useful, accessible, and transparent credential risk assessment tool. It is the sincere hope of the author that this work contributes to the broader effort to empower individuals and organisations to protect their digital identities in an increasingly adversarial online environment.

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